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**Clark et al.**

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(54) **GRAPE PLANT NAMED ‘GRATITUDE’**

(50) Latin Name: *Vitis labrusca* L.×*Vitis Vinifera* L.  
Varietal Denomination: **Gratitude**

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(58) **Field of Classification Search**  
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See application file for complete search history.

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(57) **ABSTRACT**

Description and specifications of a new and distinct grapevine  
cultivar named ‘Gratitude’ which originated from a hand-  
pollinated cross of A-1925 (non-patented, non-released  
breeding genotype; female)×A-1581 (non-patented, non-re-  
leased breeding genotype; male). This new grapevine cultivar  
can be distinguished by its seedless, green, crisp berries with  
neutral flavor and resists cracking from rainfall at maturity.  
The fruit grows in large tight clusters, the vines have medium  
vigor, and the plants are healthy.

**3 Drawing Sheets**

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Latin name: *Vitis labrusca* L.×*Vitis vinifera* L.  
Varietal denomination: ‘Gratitude’.

**BACKGROUND**

The new and distinct cultivar of grape named ‘Gratitude’ is  
described herein. The new cultivar originated from a hand-  
pollinated cross of A-1925 (female parent) and A-1581 (male  
parent) made in 1991. The seedlings fruited in the summer of  
1992 in a vineyard near Clarksville, Ark. and one was selected  
for its seedless, green, crisp berries with neutral flavor. The  
fruit grows in large tight clusters, the vines have medium  
vigor and the plants are healthy.

**SUMMARY OF THE INVENTION**

The new and distinct cultivar of grapevine originated from  
a hand-pollinated cross of A-1925 (non-patented, non-re-  
leased genotype; female parent)×A-1581 (non-patented, non-  
released breeding genotype; male parent) made in 1991 near  
Clarksville, Ark. The instant cultivar is a hybrid of *Vitis labr-*  
*usca* L. and *Vitis vinifera* L. The seeds resulting from this  
controlled hybridization were germinated in a greenhouse  
during the winter of 1991-92. Resulting seedlings were  
planted in the spring of 1992 in a vineyard near Clarksville,

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Ark. The seedlings fruited in the summer of 1994 and one  
seedling, designated Arkansas Selection 2505, was selected  
for its seedless, green, crisp berries with neutral flavor and  
resistance to fruit cracking in rainfall at maturity. The fruit  
grows in large tight clusters, the vines have medium vigor and  
the plants are healthy.

During late 1994 and early 1995, the original plant selec-  
tion was propagated asexually at the above-noted location, by  
rooting hardwood cuttings and a test planting of three vines  
was established. In all propagations hardwood cuttings were  
used and the instant cultivar rooted readily from hardwood  
cuttings. All propagules (resulting plants) of the instant cul-  
tivar have been observed to be true to type in that during all  
asexual multiplication, the vegetative and fruit characteristics  
of the original plant have been maintained. All vines planted  
from hardwood cutting propagation fruited in the second or  
third season of growth in the vineyard after planting.

Vines of the new cultivar have medium vigor, with a proc-  
umbent growth habit characteristic of *V. labrusca*. It has pro-  
duced well as own-rooted plants in all testing and has not been  
evaluated on any rootstocks. Hardiness of the vines has been  
very good, although slight winter injury has occurred on the  
vines when winter lows of 5° F. occurred but injury was not  
seen at winter temperatures above this level.



The new cultivar is moderately resistant to powdery mildew (*Erysiphe necator* Schw. (syns. *Uncinula necator* (Schw.) Burr., *E. tuckeri* Berk., *U. americana* Howe, and *U. spiralis* Berk. & Curt; anamorph *Oidium tuckeri* Berk.), downy mildew (*Plasmopora viticola* Berl. & Tomi.), and anthracnose (*Elsinoe ampelina* (d. By.) Sher), but susceptible to black rot (*Guignardia bidwellii* (Ell.) V. & R.). Fungal diseases can be controlled by the use of available fungicides.

The new cultivar ripens its fruit in the late season, from August 19-28. The fruit is green in color at early maturity. It is evenly colored within the cluster. The fruit shape is elongated oval. Fruit skins are medium-thick and adhere to the flesh (has a non-slip-skin texture). The berries are medium (ca. 3.6 g). The flavor is neutral in character. Solids concentration of the juice at fruit maturity averages 19% with medium acidity. The fruit is of the stenopermocarptic type of seedlessness and usually contains no seed traces. Fruit clusters, borne usually one to two per shoot, are large and tightly filled with an average weight of 516 g. The fruit clusters are considered large in size, good for commercial markets, and the variety is also intended for home garden planting.

The new cultivar has been named the 'Gratitude' cultivar.

#### BRIEF DESCRIPTION OF THE PHOTOGRAPHS

The accompanying photographs show typical specimens of the new variety at eight years of age in color as nearly true as it is reasonably possible to make in a color illustration of this character.

FIG. 1 is a photograph showing typical specimens of the fruit.

FIG. 2 is a photograph showing the leaf adaxial view.

FIG. 3 is a photograph showing the leaf abaxial view.

#### DETAILED DESCRIPTION OF THE NEW CULTIVAR

'Gratitude' differs from its female parent A-1925 in that it is seedless and has green berry color compared to blue color for A-1925. 'Gratitude' differs from its male parent A-1581 as this parent is blue/black in fruit color, and has much smaller clusters.

The following is a detailed description of the botanical and pomological characteristics of the subject grapevine. Color data are presented in Royal Horticultural Society Colour Chart designations, 1986 version, second edition.

Where dimensions, sizes, colors and other characteristics are given, it is to be understood that such characteristics are approximations of averages set forth as accurately as practicable.

The descriptions reported herein are from specimens grown near Clarksville, Ark. Vines used for measurement were irrigated using trickle (drip) irrigation. The data collection was from vines that were 15 years old.

Vine:

*Size*.—Medium.

*Growth*.—Moderately vigorous.

*Density of foliage*.—Medium.

*Productivity*.—Moderately productive.

*Rootstock*.—None; vines tested were own-rooted vines.

*Cold hardiness*.—Slight winter injury has occurred on the vines when winter lows of 5° F. occurred but injury was not seen at winter temperatures above this level.

*Shoots (current-season canes)*.—Color of shoots on the side exposed to direct sunlight — Red-Purple Group

(59A). Color on the side shaded from the sun Yellow-Green Group (146C). Anthocyanin present on the shoot sides exposed to direct sunlight. Shoot attitude is procumbent.

*Canes (mature measured in winter)*.—Color of mature cane: base Greyed-Orange Group (165A); midpoint Greyed Orange Group (165A); terminal Greyed-Orange Group (165A); anthocyanin not observed on mature canes at base, midpoint, or terminal. Length average 2.5 m with range of 1.1 to 3.4 m. Diameter of mature cane: base 1.1 cm, midpoint 0.7 cm, terminal 0.3 cm. Internode length: base 4.7 cm, midpoint 7.5 cm, terminal 6.5 cm. Lenticels present and dense on mature canes and small (less than 0.5 mm in diameter). Canes mature to tips in the fall.

Trunk:

*Shape*.—Slender.

*Trunk straps*.—Long, split.

*Surface texture*.—Shaggy.

*Inner bark color*.—Greyed-Orange Group (166A).

*Outer bark color*.—Grey Group (201A).

Foliage:

*Leaves*.—Leaves simple and alternate; shape orbicular; number of lobes 1; petiole sinus shape open to half open; venation palmate; margin serrated with shape of teeth convex and teeth medium in size. Pubescence is sparse on mature leaves. Color of mature leaves: base abaxial — Yellow-Green Group (146B); base adaxial — Green Group (137B); midpoint abaxial — Yellow-Green Group (146B); midpoint adaxial — Green Group (137B); terminal abaxial — Yellow-Green Group (146B); terminal adaxial — Green Group (137B). Anthocyanin present lightly on the main and secondary veins. Color of young leaves: base abaxial — Yellow-Green Group (146C); base adaxial — Yellow-Green Group (146B); midpoint abaxial — Yellow-Green Group (146D); midpoint adaxial — Yellow-Green Group (146C); terminal abaxial — Yellow-Green Group (148D); terminal adaxial — Yellow-Green Group (146C).

*Petioles*.—Color on young leaves — Yellow-Green Group (146D). Color of mature petioles: abaxial side Yellow-Green Group (146D), adaxial side Greyed-Red Group (182B). Petiole anthocyanin present on the surface where it is most exposed to sunlight. Sinus of mature leaf is 4.9 cm deep and 6.0 cm at widest point. Mature leaves have sparse pubescence on midrib, main veins and secondary veins. Young leaves have medium pubescence on midrib, and sparse pubescence on main veins and secondary veins.

*Tendrils*.—Intermittent and opposite the leaf node. Length 25.8 cm. Texture smooth and usually forked and curled on distal end. Color of mature tendril Yellow-Green Group (146C).

*Buds*.—Average number of buds on a current, single-season cane 38. Dormant bud (compound bud or eye) width 3.4 mm; shape triangular and slightly rounded. Color Greyed-Orange Group (165A). Texture smooth.

Disease resistance: Moderately resistant to powdery mildew, and downy mildew; susceptible to black rot. Other disease or pest susceptibilities not known.



Flowers:

*Sex.*—Hermaphrodite.  
*Date of bloom.*—May 11 (first); May 16 (full). Flowers  
per cluster — 341.  
*Stamens.*—Number: 5. Color: Green-White Group 5  
(157D).  
*Pistil.*—Number: 1. Length: 3.0 mm. Color: Yellow-  
Green Group (145A).  
*Pollen.*—Color: Yellow Group (13C), normal and fer-  
tile. Petal — Cap of 5 fused petals, flat and round. 10  
Color — Yellow-Green group (147C).  
*Sepal.*—None.

Fruit:

*Maturity.*—Late season August 19-28.  
*Berry.*—Shape — Elongated oval. Color — Yellow-  
Green Group (146C). Size — Diameter at equator: 1.6 15  
cm. Diameter at base: 1.4 cm. Diameter at apex: 1.3  
cm. Length: 2.4 cm. Weight: 3.6 g; uniform in size.  
Texture — Non-slipskin. Skin thickness — Medium.  
Seeds — Seedless with 0 seeds per berry. Brush  
length — 5.48 mm. Flavor — Sweet, neutral flavor. 20  
Soluble solids — 16.2%.

*Juice.*—pH — 3.7. Titratable acidity — 2.7 g/L tartaric  
acid.  
*Cluster.*—Weight — 516 g. Length — 14.8 cm.  
Width — 5.0 cm. Berries per cluster — 140-254,  
mean=207.4. Cluster per vine — 47.4. Clusters per  
shoot — One to two. Peduncle length — 1.91 cm.  
Pedicle: Length — 1.50 cm. Diameter — 0.12 cm.  
Color — Yellow-Green Group (146D). Use — Fresh  
consumption as a table grape is the primary use par-  
ticularly for local markets. No processing evaluations  
done.

The cultivar: The most distinctive features of the cultivar are  
its seedless, crisp, green, thin-skinned berries with neutral  
flavor and resistance to fruit cracking from rainfall at matu-  
rity, large, tight clusters, medium vigor, and healthy plant.

We claim:

1. A new and distinct cultivar of grape plant named ‘Grati-  
tude’ substantially as illustrated and described.

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Figure 1





Figure 2

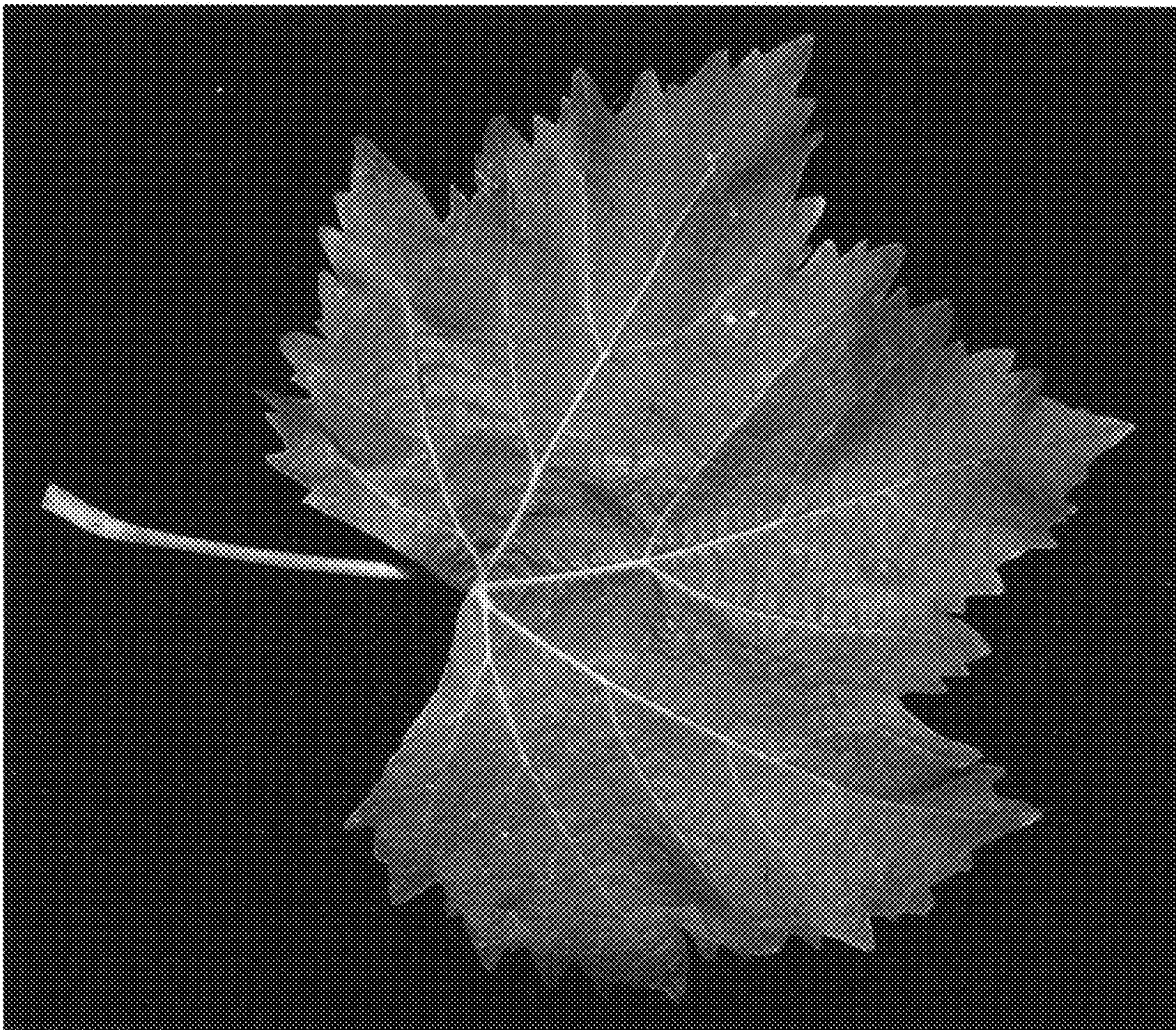




Figure 3

